Category Correlation and Adaptive Knowledge Distillation for Compact Cloud Detection in Remote Sensing Images

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***Abstract*— Cloud detection relying on deep convolutional neural networks (DCNNs) obtains remarkable accuracy gains at the expense of high computation and storage costs, which are difficult to deploy to resource-constrained devices, such as intelligent satellites. Recently, knowledge distillation (KD) has been a promising solution for a compact model. However, most existing KD methods only transfer the feature relationship of pairwise pixel, which fails to cope with thin clouds and cloud-like objects in complex scenes. Furthermore, those KD methods directly imitate the output of the complicated model regardless of the correctness. In this article, we propose a novel category correlation and adaptive KD (CAKD) framework for the lightweight cloud detection network. We design a category relational context (CRC) module to refine the structured pixel-category correlation from the teacher and student network. Then, we perform the category correlation distillation (CCD) to make the student model better address the intraclass consistency and the interclass difference, thus reducing the category confusion. Besides, a pixel-adaptive distillation (PAD) module is utilized to adaptively transfer the soft-output knowledge of a teacher model by extracting the teacher's pixel prediction probability. Extensive experiments on Landsat 8, Landsat 7, Gaofen-2, Gaofen-1, and Google Earth dataset report the effectiveness and universality of our distillation method. The CAKD allows MobileNetV2 with 2.31M parameters and 4.63G float-point operations (FLOPs) to outperform advanced cloud detection methods without the added inference overhead.**

***Index Terms*— Cloud detection, compact model, knowledge distillation (KD), relational context, remote sensing images (RSIs).**

I. Introduction

DEEP neural networks have recently been implemented for high-precision cloud detection on remote sensing images (RSIs) contaminated by atmospheric clouds [1]–[9]. With the development of intelligent satellites, onboard cloud detection is trending as a promising technology. For example, the Qilu-2 satellite is equipped with modular computers to implement the edge device deployment of a cloud detection model based on deep learning. The low-value cloudy images can be eliminated in advance by onboard cloud detection, thus improving the utilization rate of RSIs and the data transmission efficiency of satellites. However, most current cloud detection networks are difficult to deploy on satellites with limited storage memory and computing resources. Therefore, there is an urgency for developing the lightweight and accurate cloud detection model in practical applications.

Several approaches have been proposed to implement model compression. A straight way is to compress the cumbersome network through pruning [10], [11], quantization [12], [13], or decomposition [14], [15], resulting in a compressed model with fewer parameters and computations. Another way designs specific convolution to create a lightweight but powerful network, e.g., MobileNetV2 [16], ShuffleNet [17], and so on. However, these methods often sacrifice superior performance, and there is a clear gap in estimation accuracy between the compressed and complex models. Recently, knowledge distillation (KD) [18] is proven to be a promising solution for maintaining the performance of compact models with its simplicity and efficiency. KD allows the lightweight student network to imitate the constructive knowledge of the sophisticated teacher network, boosting the generalization ability of the student network.

Following KD for the image classification [18]–[22], the exploration of KD on dense semantic segmentation has drawn increasing attention. Cloud detection labeling clouds in images can be treated as a semantic segmentation task. There have been many impressive results relying on the teacher–student KD for semantic segmentation. He *et al.* [23] combined a pre-trained feature encoder to distill the affinity between any two positions. Liu *et al.* [24] proposed pairwise distillation, which transfers pairwise relations in the spatial position of the feature. Feng *et al.* [25] exploited cross-layer spatial attention to distill pixelwise similarity. The abovementioned methods have been proven effective for training compact semantic segmentation networks in natural scenes.

Unlike objects in natural scenes, clouds with irregular forms are more challenging, and the following facts manifest the dilemma of cloud detection. On the one hand, thin clouds in RSIs appear translucent, causing the coverage area of thin

Manuscript received 1 March 2022; revised 24 April 2022; accepted 10 May 2022. Date of publication 13 May 2022; date of current version 26 May 2022. This work was supported by the National Natural Science Foundation under Grant 62171436. *(Corresponding author: Zhiyuan Yan.)*

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Digital Object Identifier 10.1109/TGRS.2022.3174910

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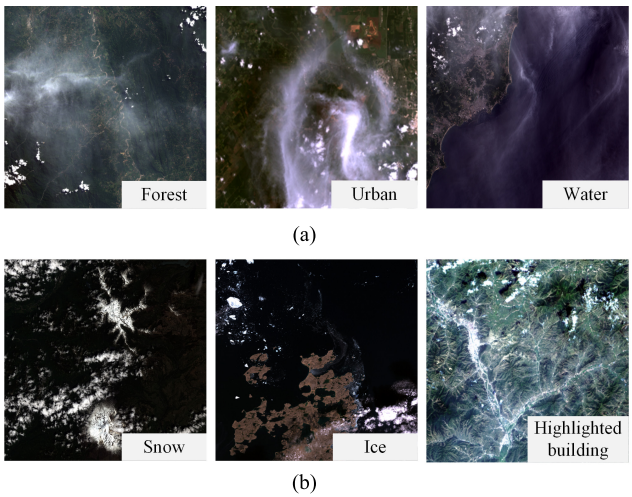

Fig. 1. Examples of dilemmas in cloud detection. (a) Translucent thin clouds with some background information have complex and varied target characteristics in different scenes. (b) Cloud-like objects in the background are often misclassified as clouds.

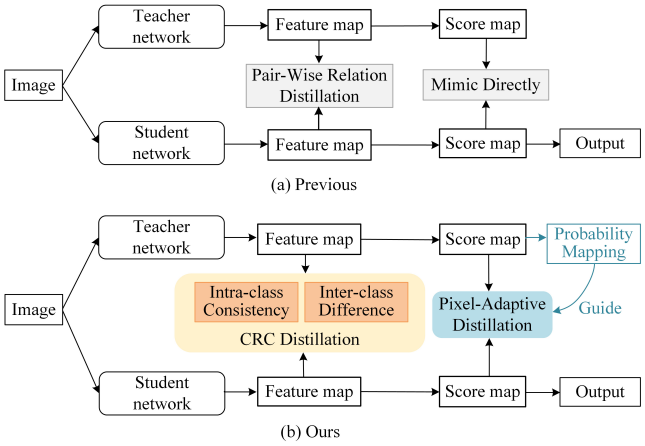

Fig. 2. Pipelines of KD in cloud detection. (a) Previous method only calculated the pairwise relationships ignoring the category information of the pixels. The student network directly imitates the score map of the teacher. (b) Proposed CAKD distillation framework. The CRC distillation transfers intraclass and interclass relationships of the feature map. The PAD extracts the probability values of the teacher for adaptive distillation.

clouds to contain some information from ground objects. As shown in Fig. 1(a), influenced by different scenarios, e.g., forest, urban, and water region, thin clouds usually present complicated and diverse target characteristics. Consequently, to learn more accurate feature representation, the lightweight network should enhance the intraclass feature consistency of thin clouds to capture the common feature in different contexts. On the other hand, cloud-like objects, such as snow, ice, and highlighted objects, as shown in Fig. 1(b), interfere with cloud detection and are often misclassified as clouds. The student network tends to engender feature confusion between clouds and cloud-like objects in such cases. Therefore, it is crucial for the lightweight network to boost the feature difference between categories and, thus, distinguish confusable targets. Nevertheless, the existing KD approaches use mainly the spatial pixel-by-pixel relationships of intermediate features without considering the category dependencies of pixels in the feature map, which results in poor performance. Although a bidirectional self-distillation method [26] has been proposed for compact cloud detection, but the compact model with low abstraction capacity inherently struggles to capture thin clouds in complex scenarios. Overall, exploring the category correlation of the feature is a conducive proposal in the distillation of compact cloud detection.

In addition, current distillation methods always perform pixel-level direct imitation of the teacher's score map. However, the teacher network may not always give the correct guidance as the ground truth, especially at the irregular and complex edge of clouds. Misclassified pixels of the teacher network will lead the student network to update the gradient with the wrong soft label. Accordingly, it is desirable that the teacher network instructs student models adaptively with its own discriminative ability.

To overcome these challenges in cloud detection, we propose a novel category correlation and adaptive KD (CAKD) approach for lightweight cloud detection model, as shown in Fig. 2. We design a category relational context (CRC) module in the feature map of the networks. The CRC module calculates the similarity distribution for each pixel and category feature, which embodies the pixel-category correlation of the feature. Afterward, the CRC knowledge, i.e., the consistency of features belonging to the same category and the variability between different categories, is transferred through category correlation distillation (CCD). It allows the student network to learn rich and powerful categorical structured knowledge and enhance the robustness of cloud detection under the influence of complex backgrounds. Besides, a pixel-adaptive distillation (PAD) module assists in adaptively distilling the score map of the teacher network. The PAD module guides distillation with a pixel-level predicted probability mapping of the teacher network, thus avoiding the erroneous knowledge from the teacher network.

In summary, the contributions of our work are summarized as follows.

1) We present an effective distillation framework called CAKD for the lightweight and accurate cloud detection model. Compared with other cloud detection methods, our distillation method enables the model to achieve optimal performance with fewer parameters and computational complexity.
2) The CRC module is proposed to construct category relationship of the feature map. Through CCD, the student captures the common feature representation of thin clouds and avoids the confusion from cloud-like objects.
3) The PAD module adaptively transfers the score map of the teacher network to avoid negative soft-output knowledge. The distillation loss is adaptively weighted based on the predicted confidence of teacher network.

The remainder of this article is organized as follows. We describe related works of our approach in Section II.

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Section III illustrates the proposed distillation framework. Section IV provides the experiments and the discussion. Finally, we summarize our work in Section V.

II. Related Works

We introduce related works involved in our research in this section. We focus on three aspects: cloud detection methods, KD for model compression, and relational context information.

A. Cloud Detection Methods

Recently, cloud detection is an important pre-processing stage with the widespread application of RSIs. Traditional cloud detection methods consist of two main categories: threshold methods based on physical characteristics [27]–[31] and methods based on machine learning [32]–[34]. The threshold method is usually designed for hyperspectral image data. For example, the cloud detection method used for Moderate-Resolution Imaging Spectroradiometer (MODIS) data [27] selects 14 bands to set different thresholds in different scenes (such as day or night, ocean, snow, and so on) for discrimination. The Fmask algorithm [29] is widely used in Landsat satellite data. By selecting seven spectral bands and combining NDVI and NDSI, Fmask uses several thresholds for cloud detection. The threshold-based methods are simple and effective, but they heavily rely on the spectral characteristics, which lacks universality. In addition, some machine learning methods, including classifier-based supervised learning methods, such as support vector machine (SVM) [32] and random forest (RF) [33], and clustering algorithms, such as K-Means [34], are applied to cloud detection. These machine learning-based methods usually need to select appropriate features, such as spectral or physical features, texture features, edge features, and so on, which often cannot adapt to complex scenarios.

In recent years, inspired by deep learning in computer vision, many studies [1]–[9] regard cloud detection as an image semantic segmentation task and design specific deep convolutional neural network (DCNN) to achieve high-precision results. Shao *et al.* [4] proposed a multiscale feature module for thin and thick cloud detection with the combination of multispectral images as input. Yang *et al.* [5] and Guo *et al.* [7] designed CDNet and CDNetv2 successively to detect clouds on challenging thumbnails. Among them, different optimization modules are proposed for enhancing network performance. Li *et al.* [9] treated cloud detection as the problem of energy separation between clouds and backgrounds and extracted the cloud mask using the deep matting technique. Although these DCNN-based approaches achieve a breakthrough in performance, the enormous network models have a huge amount of parameters and computations. For the practical application of edge devices, such as intelligent satellites, finding a trade-off between accuracy and efficiency is a vital challenge that remains to be tackled.

Some lightweight cloud detection models have been proposed recently. Zhang *et al.* [35] combined U-net with image compression to design a lightweight cloud detection model for small satellites. Chai *et al.* [26] presented a bidirectional self-attention distillation and enhanced the edge and interior representations of clouds. The distillation method was effectively used for Gaofen-1 RSIs through bidirectional layerwise feature learning. He *et al.* [36] used ShuffleNet as the lightweight backbone, thus decreasing the parameters and float-point operations (FLOPs) of the model. Yet, these methods still fail to capture thin clouds in complex remote sensing scenes accurately.

B. KD for Model Compression

Previous methods of model compression aim to create a small but powerful network directly. One type of method is network pruning [10], [11], quantization [12], [13], or decomposition [14], [15], and another is to design specific convolution structures to replace the original convolution [16], [17], [37], such as the depthwise separable convolution in MobileNetV2 [16]. With reduced parameters and computations, these methods effectively adapt to real-time inference on mobile devices. However, these methods sacrifice the network's performance and cannot yield prediction results comparable to the complicated network. KD is first introduced by Hinton *et al.* [18] and has drawn much attention in model compression studies. With the teacher–student KD framework, the intrinsic information learned by a complex teacher network is transferred to a compact student network, thereby enhancing the performance of the compact network. For example, Park *et al.* [19] proposed relational KD to transfer the structural relationship information between samples. FitNet [20] and attention transfer (AT) [21] distill the intermediate representation and the attention mapping from the feature map of the teacher network, respectively. Yim *et al.* [22] proposed an approach based on the flow of solution procedure, which calculates the Gramian matrix between the feature layers as a carrier of teacher network knowledge.

Following the success of KD in the classification task, KD has recently been implemented for the compact semantic segmentation network in natural scenes. He *et al.* [23] encoded the teacher's feature output as an easily learned compact knowledge and transferred the affinity relations for any two positions. Liu *et al.* [24] proposed pairwise distillation of the feature map, which builds the spatial relational graph of pixels with their neighborhoods or even the whole image. An extra adversarial network is used for training the intrinsic coherence of the score map. Feng *et al.* [25] exploited the pixelwise similarity by subtracting the cross-layer attention map, with less calculation cost. Wang *et al.* [38] proposed to distill intraclass feature variation, which calculates the feature similarity between a pixel and the class it belongs to. The abovementioned distillation methods regulate the intermediate feature maps of the network, on which the spatial relational context is distilled. Nevertheless, the dense pixel-by-pixel relationship neglects the class attributes of the pixel features. More importantly, cloud detection faces the dilemma of category ambiguity caused by thin clouds and cloud-like objects, which affects cloud detection. The consistency and discrepancy information of the feature map is more significant for cloud detection.

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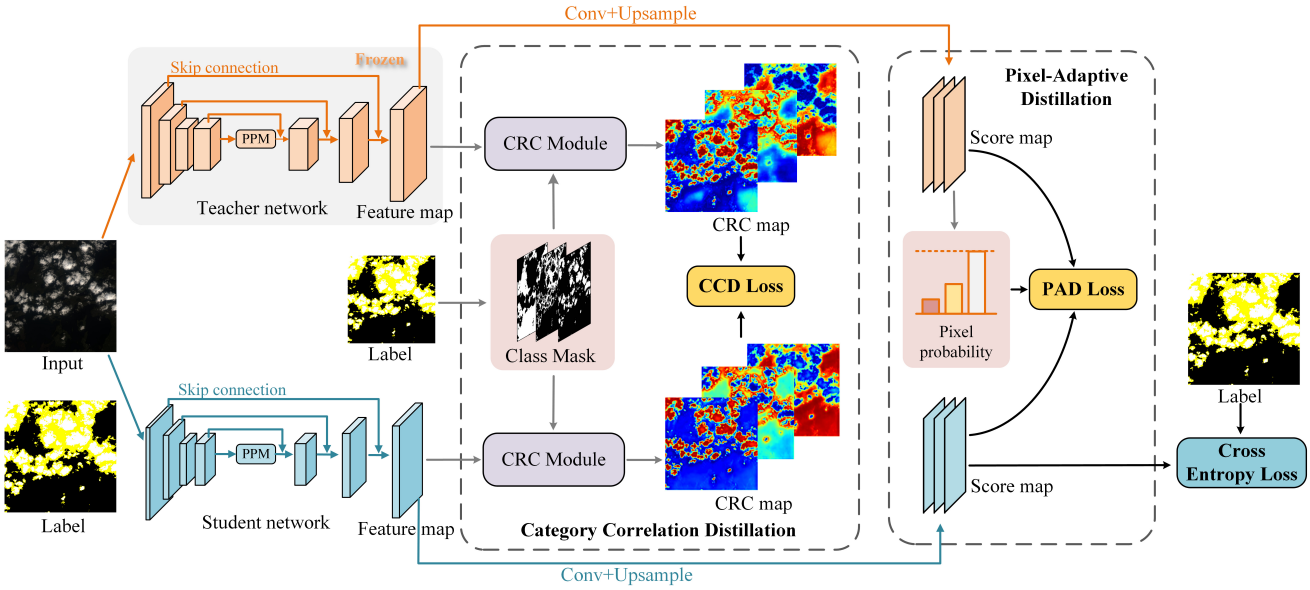

Fig. 3. Overall architecture of our CAKD framework. The student network is not only supervised from the labels, but also constrained by the knowledge from the teacher network. The CCD transfers the CRC of feature map refined by the CRC module. The PAD adaptively imitates the soft output of the teacher.

C. Relational Context Information

As with the semantic segmentation task, exploring context information is crucial for the cloud detection network. Aggregating context information to capture rich pixel dependencies has been widely applied in semantic segmentation. Previous works, such as pyramid pooling module (PPM) [39] and atrous spatial pyramid pooling (ASPP) [40], exploit the spatial multiscale context information. In recent years, many segmentation networks have considered the relational context. The calculation of the relational context is divided into pixelwise and pixel-region interactions. For the former, some networks [41]–[43] consider the pixelwise similarity and use the similarity as the weight to aggregate the feature context. For instance, DANet [41] captures global feature dependencies in the spatial and channel dimensions and selectively aggregates feature contexts. There are also several segmentation networks that utilize the context of the pixel-region relationship. Zhang *et al.* [44] proposed ACFNet to integrate global contexts based on class regions and calculate the attentional class feature of pixels by the coarse segmentation result. Yuan *et al.* [45] proposed OCRNet, which calculates the similarity between object region representations and pixel features and adds an object context representation for each pixel.

Some progress has also been made in applying relational context to KD. The pairwise distillation in [24] calculates the feature affinity map for each pixel. The intraclass feature variation [38] is described as the relationship between pixels and the corresponding category area feature. However, such pixel-level relational context has limitations for clouds that are easily confused between categories. Different from previous works, we comprehensively build the intraclass and interclass relational context of the intermediate feature map in the distillation.

III. Methodology

In this section, we first describe the proposed CAKD framework, and two distillation modules are presented, respectively. Finally, we introduce the training process of our distillation framework.

A. Overview

KD works by transmitting the constructive information of the well-performed teacher network to the compact student network, improving the cloud detection performance. For input images $X=\{x_1, x_2, \ldots, x_N\}$, our framework is intended for the student network with parameters θ_s to mimic the behavior of the teacher defined as T. With the perspective of maximum likelihood, the optimization problem of the KD task can be expressed as

$$\begin{aligned}\arg\max_{\theta_s} q(T|X,\theta_s) &= \arg\max_{\theta_s} \prod_{i=1}^{N} q(t_i|x_i,\theta_s) \\ &= \arg\max_{\theta_s} \frac{1}{N}\sum_{i=1}^{N} \log q(t_i|x_i,\theta_s). \quad (1)\end{aligned}$$

The teacher network is pre-trained and fixed, assuming that the joint probability distribution of the input image x and the corresponding teacher behavior t is $p(x,t)$. Based on the law of large numbers, (1) can be approximately converted to

$$\begin{aligned}&\arg\max_{\theta_s} E_{x,t\sim p(x,t)}\left[\log q(t|x,\theta_s)\right] \\ &= \arg\max_{\theta_s} E_{x\sim p(x)}\left[\int p(t|x)\log q(t|x,\theta_s)dt\right] \\ &= \arg\min_{\theta_s} E_{x\sim p(x)}[KL(p(t|x)||q(t|x,\theta_s))] \quad (2)\end{aligned}$$

where $p(x)$ means the distribution of input images X, and KL denotes the Kullback–Leibler divergence.

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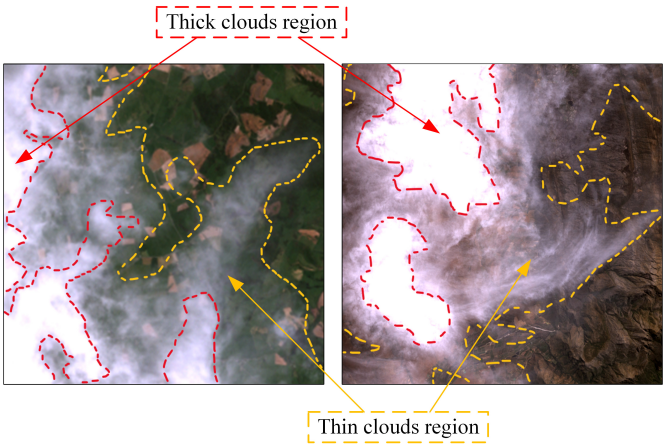

Fig. 4. Illustration of the region of thick clouds and thin clouds in RSIs. Thick clouds show the uniform highlighted color characteristic, and thin clouds present a blurred vision. Red and yellow dashed lines depict thick clouds and thin clouds areas, respectively.

Therefore, we establish the objective of KD based on KL divergence to help the student imitate the behavior of the teacher as

$$\arg\min_{\theta_s} \sum_{i=1}^{N} [KL(p(t_i|x_i)||q(t_i|x_i,\theta_s))]. \tag{3}$$

Based on the abovementioned representation, we design an effective teacher–student distillation framework called CAKD for a compact cloud detection model, as shown in Fig. 3. When training the compact network, the weight of teacher is frozen without gradient propagation. We consider the constructive knowledge from the feature map A and the soft output O to define the teacher's behavior T, denoted as

$$T \in \{G(A^t), O^t\} \tag{4}$$

where $G(\cdot)$ is conceived to extract structured information from the feature map A. In particular, the CRC module is designed to construct intraclass consistency and interclass differences of the feature of clouds. The CCD transfers the CRC map from the teacher to the student, resulting in more compact category descriptions of thin clouds and mitigating feature confusion between clouds and cloud-like objects for the student network. The PAD module extracts each pixel probability value of the teacher to guide the distillation of the soft output (score map) in an adaptive manner. Detailed operations will be described separately in Sections III-B–III-D.

B. Feature CCD

In RSIs, the area covered by thin translucent clouds can be interpreted as the energy synthesis reflected from cloud layers and the ground [9]. The lack of the compact student network's feature deteriorates the consistency of thin cloud features. Also, the thin cloud features studied by the student are similar to the background, making thin cloud pixels less detectable. Also, the appearance of cloud-like objects can cause confusion about the detailed characterization of the cloud target. The existing distillation methods have made some progress in refining knowledge on spatial location relationships. They calculate pairwise relationships in terms of pixels with surrounding neighboring pixels or even with all pixels on the full feature map. Such methods are straightforward for spatial correlation establishment of feature maps. However, they do not explicitly consider which category a pixel comes from when calculating relative relationships. The network is required to recover the category information of the pixel given the feature representation being downsampled; thus, the intrinsic link between the pixel feature and the category is essential.

As shown in Fig. 4, it is observed that in RSIs, thick clouds show uniform white and the highlighted color characteristic, while thin clouds affect the observation with the blurred visual representation. The neural network abstracts the feature description of clouds under such visual uniformity of clouds so as to segment different clouds. By assigning the correct class attribute to feature regions with the ground truth, the class prototypes (class centers) [44] are computed to unify the feature belonging to the same class of pixels. To exploit the feature category correlation of cloud features refined by the network, we propose the CRC module. As shown in Fig. 5, the CRC module measures the similarity between pixel features and different category prototypes, and the further obtained similarity distribution map about the categories depicts the category correlation of the feature map.

First, we conduct the extraction of class centers for the same-class features, i.e., class prototypes. Given the feature map $A \in R^{C\times H\times W}$, where C, H, and W denote the number of channel, height, and width, respectively. To match the same-class regions, we downsample the ground truth G to the same size as the feature map A to get masked region features belonging to different categories, and the category regions are denoted as R. The general perception is that the network maintains a consistent semantic description of the input images in the same-class regions. As such, we rely on category regions using the average pooling to aggregate features from the same category. With n category prototypes, the feature center representation of the ith category $p_i \in R^{C\times 1\times 1}$ is denoted as

$$p_i = \frac{1}{\text{num}(R_i)} \sum_{a_j \in R_i} a_j, \quad i = \{1, \ldots, n\} \tag{5}$$

where $\text{num}(R_i)$ denotes the number of pixels belonging to the ith category.

Then, we calculate the cosine similarity $S_i \in R^{H\times W}$ between the class prototype p_i and the feature $a_j \in A$ for each pixel j. $s_{i,j} \in S_i$ is denoted as

$$s_{i,j} = \text{sim}(a_j, p_i) = \frac{a_j^{\mathrm{T}} p_i}{||a_j||_2 ||p_i||_2} \tag{6}$$

where $a_j \in A$, $j \in [1, 2, \ldots, H \times W]$ is the jth location in A. Thus, S_i implies the feature pixel-region similarity map of the ith class.

Finally, the feature pixel-region similarity $S \in R^{n\times H\times W}$ is generated by incorporating the similarity maps that belong to the n class prototypes. $s_j \in S$ is denoted as

$$s_j = \text{cat}(s_{1,j}, \ldots, s_{n,j}) \tag{7}$$

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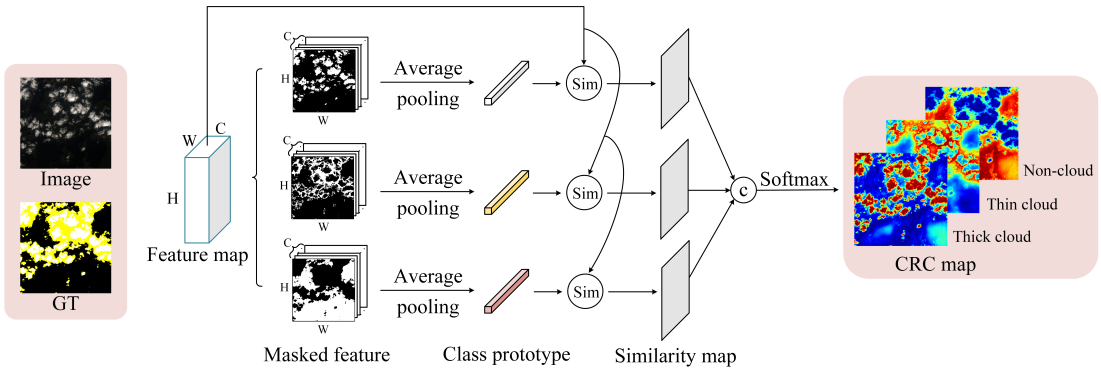

Fig. 5. Details of the CRC module. It calculates the similarity relationship between the feature map and the different class prototypes. The normalized CRC map aggregates structural knowledge both intraclass and interclass.

where cat(·) denotes a concatenation operation. To extract the universal structured knowledge of networks, we proceed to apply the softmax function along the channel dimension to obtain the normalized feature pixel-region similarity, and the formula is as follows:

$$m_{ij} = \frac{\exp(s_{ij})}{\sum_{i=0}^{n-1} \exp(s_{ij})} \tag{8}$$

where $m_{ij} \in M$. $M \in R^{n \times H \times W}$ represents the CRC knowledge we proposed, noted as CRC map.

In contrast to the previous approach [24], [38] that treat relationship measurements directly as knowledge, we normalize the similarity for pixels and class prototypes as a probability distribution. Due to the difference in depth of feature maps, directly fitting similarity values is an inappropriate guidance for the student network with relatively shallow feature depth. Structured knowledge from the feature map needs to be mined, which is expected to be consistent across feature maps of different networks. We verify the validity of our innovation in Section IV.

In Fig. 6, we extract and visualize the CRC of thin and thick clouds, respectively. The top RSI shows the typical presence of thin clouds. Thin cloud pixels are ambiguous, leading the student network high similarity in some background regions. In contrast, the teacher network has a more homogeneous category description of thin clouds. We visualize the CRC description for thick clouds for the scene with the cloud-like backgrounds (snow) at the bottom. The teacher network has a relatively more robust ability to discriminate confusable backgrounds with lower similarity in the snow region. Consequently, we match the CRC map between networks to achieve CCD, optimizing intraclass consistency and interclass differences of the student network, as shown in Fig. 3.

To transfer the CRC knowledge better, the KL divergence is used to calculate the loss between the CRC maps of the networks. The CCD loss L_{CCD} is as follows:

$$\begin{aligned} L_{\mathrm{CCD}} &= KL\left(M^s \parallel M^t\right) \\ &= \frac{1}{H \times W} \sum_{j=1}^{H \times W} \sum_{i=1}^{n} M_{j,i}^s \log\left(\frac{M_{j,i}^s}{M_{j,i}^t}\right) \end{aligned} \tag{9}$$

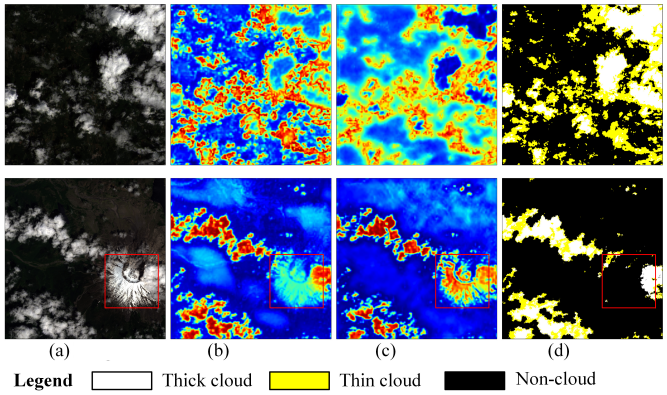

Fig. 6. Visualization of CRC. The top represents the CRC for thin clouds. The bottom represents the CRC for thick clouds, and the red box indicates the snow region. (a) Image. (b) Teacher. (c) Student. (d) Label.

where $M_{j,i}^t$ and $M_{j,i}^s$ denote the CRC between the jth pixel in feature map A and the ith category prototype of the teacher and the student network, respectively.

C. Pixel-Adaptive Distillation

The CCD accomplishes representation learning on the high-level embedding space. Just as distillation was proposed initially, the softmax output of the teacher network contains the information of negative samples, which contributes to the generalization capability of the student network. Likewise, semantic segmentation is commonly treated as a pixel-level classification task, with KD carrying out pixel-level knowledge imitation.

Nevertheless, the teacher network has the variable predictive capability for different pixels for complex remote sensing scenarios. The injection of the correct predictive distribution is positive for the training of the student network. But, for difficult pixels, especially at irregular and complex edges of clouds, the teacher network may not always provide the correct guidance compared with the label. As shown in Fig. 7, the teacher network could predict correctly with a slight probability margin in the green box or even give guidance that is negative in the red box. Thus, we propose PAD loss with adaptive teaching to avoid local pixel misdirection.

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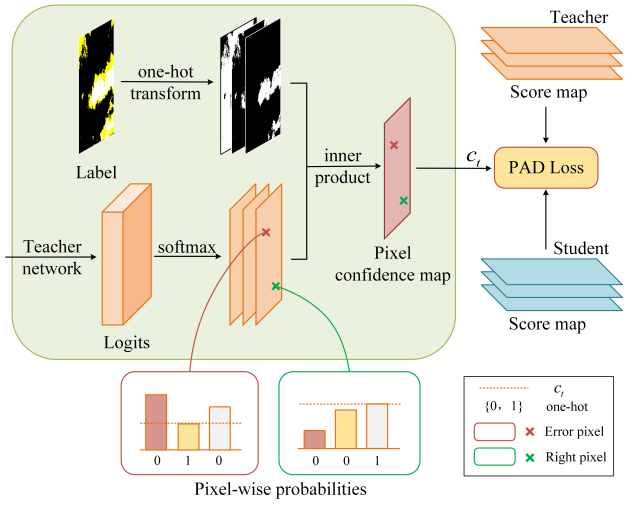

Fig. 7. Illustration of the PAD module. The pixel prediction probabilities of the teacher network are extracted to weigh the distillation loss.

The adaptive distillation process is described as follows. Assuming pixel-level logits z for the network, the probability p_i belonging to the ith class is calculated using the softmax function. In addition, the temperature parameter τ is adjusted to soften the probability distribution, and the formula is

$$p_i = \frac{\exp(z_i/\tau)}{\sum_j \exp(z_j/\tau)}. \tag{10}$$

For dense prediction tasks, larger τ introduces noise, which is not helpful for learning. In our study, τ is set to 1.

Fig. 7 shows our PAD module. Given the teacher's output score map O by softmax and the label G, we extract the pixel confidence map C_t by the pixel-by-pixel inner product operation $C_t = \text{Inner}(O^t, T_{\text{OneHot}}(G))$. $c_j \in C_t$ for each pixel j is denoted as

$$c_j = p_{j,(i|i=G_j)} \tag{11}$$

in which G_j is the label of the jth pixel. In other words, the value of c_j is equal to the probability value of the ith class in the probability distribution p_j.

Subsequently, we introduce KL divergence to implement the distillation of score maps, which is defined as follows:

$$L_{\text{KD}}\left(p\left(z^t\right), p\left(z^s\right)\right) = \sum_i p_i\left(z_i^t\right) \log\left(\frac{p_i\left(z_i^s\right)}{p_i\left(z_i^t\right)}\right). \tag{12}$$

We take the extracted pixel confidence map as the teaching ability of the teacher. Then, we utilize the pixel prediction probability C_t to reweigh the distillation loss function, and the PAD can be denoted as

$$\begin{aligned} L_{\text{PAD}}\left(O^t, O^s\right) &= C_t * L_{\text{KD}}\left(O^t, O^s\right) \\ &= \sum_j c_j L_{\text{KD}}\left(p_j^t, p_j^s\right) \end{aligned} \tag{13}$$

where O_t and O_s denote the score maps of the teacher and student network, respectively.

Algorithm 1 Training Process for Lightweight Cloud Detection Based on Our Distillation Method

Input: Teacher network T, and its weights θ_t; The RSIs I, and the corresponding ground-truth labels G.
Output: Student network S, and its weights θ_s.

Initialize the weights of the student network θ_s;
Step 1: Feature Category Correlation Distillation
Calculate the feature map of the networks respectively, A^t and A^s;
for *class* $i \in \{0, 1, 2\}$ **do**
  Downsample the ground-truth G to match the spatial size of the feature map, get the category regions R_i;
  Calculate the category prototypes p_i^t and p_i^s;
  Similarity measure $S_i^t, S_i^s \leftarrow sim(A^t, p_i), sim(A^s, p_i)$;
  $S^t, S^s \leftarrow cat(S_i^t, S_{i+1}^t), cat(S_i^s, S_{i+1}^s)$;
end for
Normalize the similarity $M^t, M^s \leftarrow Nor(S^t), Nor(S^s)$;
Calculate the category correlation distillation loss L_{CCD};
Step 2: Pixel-Adaptive Distillation
Calculate the segmentation score map O^t and O^s;
Calculate the pixel confidence map for the teacher network $C_t \leftarrow Inner(O^t, T_{OneHot}(G))$;
Calculate the category correlation distillation loss L_{PAD};
Step 3: Update the weights of the student network
Calculate the cross-entropy loss L_{CE};
$\theta_s \leftarrow argmin(L_{CE} + \alpha L_{\text{CCD}} + \beta L_{\text{PAD}})$
return θ_s

D. Optimization Process

Based on a pre-trained teacher network, we train the compact student network, and the optimization process is shown in Algorithm 1. The CCD and PAD implement knowledge transfer at the feature and pixel level, respectively, only during training. The lightweight student network is free of extra parameters and computations in the inference phase. The overall objective function is composed of three components: the standard cross-entropy loss L_{CE}, the CCD loss L_{CCD} in (9), and the PAD loss L_{PAD} in (13)

$$\text{loss} = L_{\text{CE}} + \alpha L_{\text{CCD}} + \beta L_{\text{PAD}} \tag{14}$$

in which, α and β are loss weights making the loss value ranges comparable.

IV. Experiments

This section comprehensively evaluates our CAKD framework in multiple datasets. In particular, we introduce the datasets, implementation details, and experimental results.

A. Datasets

We conduct a series of experiments on multiple optical satellite images with different spatial resolutions, as shown in Table I. They are the publicly released Cloud Cover Assessment Validation dataset of Landsat 8 [46] and Landsat 7 [47], the AIR-CD dataset [36], HRC-WHU dataset [6], and Gaofen-1 (GF-1) wide-field-view (WFV) Cloud and Cloud Shadow Cover Validation Data [30].

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TABLE I
MAIN CLOUD DETECTION DATASETS IN THE EXPERIMENTS. *NOTE THAT THE IMAGE SIZE DENOTES THE SIZE OF THE IMAGE ACTUALLY USED IN THE EXPERIMENTS

Dataset	Source	Resolution	Class	Image size
Landsat 8	Landsat-8 OLI/TIRS	30m	thin cloud, thick cloud	~7000 × 6000 × 10*
Landsat 7	Landsat-7 ETM+	30m	thin cloud, thick cloud	~7000 × 6000 × 4*
AIR-CD	Gaofen-2 PMS	4m	cloud	7300 × 6908 × 4
HRC-WHU	Google Earth	0.5-15 m	cloud	1280 × 720 × 3
GF-1	Gaofen-1 WFV	16m	cloud, cloud shadow	~17000 × 16000 × 4

1) Landsat 8 Cloud Cover Assessment Validation Data: The Level-1 multispectral data are captured by Landsat 8 satellite Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS), including 96 scenes images with a spatial resolution of 30 m. It covers eight kinds of natural scenes, including barren, grass/crops, forest, shrubland, urban, snow/ice, wetlands, and water area. They are digitized with the categories of clear sky, thin cloud, thick cloud, and cloud shadow, in which the thin cloud is visually translucent. In our experiments, we treat the clear sky and cloud shadow as non-cloud, and there are three categories: thick clouds, thin clouds, and non-clouds. To ensure a reasonable distribution of the dataset, we select 64 images with the cloud percentage of 5%–100% and divide them into the training set, the validation set, and the test set in the proportion of 2:1:1. We slice the original images into 512 × 512 pixels in the training and testing phases. The experiments take visible bands as the input to the network. Since the dataset contains rich multispectral bands information, apart from the panchromatic band with a spatial resolution of 15 m, we also utilize the near-infrared (NIR), the shortwave infrared (SWIR), and the thermal infrared (TIR) bands, which are favorable for cloud extraction for further experiments.

2) Landsat 7 Cloud Cover Assessment Validation Data: There are 206 Landsat 7 Enhanced Thematic Mapper Plus (ETM+) Level-1G scenes. These scenes are distributed according to different latitudinal bands. To avoid intolerable mislabeling, we select 44 of these images with relatively accurate annotations for our experiments and divide them into three parts: 23, 10, and 11 images for training, validation, and testing, respectively. The scenes we chose in our experiments include barren, water, urban, snow/ice, wetlands, and vegetation. The data preprocessing and the category setting are consistent with the Landsat 8 dataset. We perform comparison experiments using bands 1–4 of this dataset, where bands 1-3 are the visible bands, and band 4 is the NIR band, which is beneficial to cloud detection.

3) AIR-CD Dataset: The AIR-CD dataset contains 34 high-resolution RSIs from the GaoFen-2 satellite with a spatial resolution of 4 m. These images are collected in different regions of China and are labeled by pixel as cloud and background. In our experiments, all 34 images are randomly divided into 25 for training and 9 for testing. The original images are sliced to a size of 512 × 512 pixels, and we take the visible bands and NIR band as the input to the network.

4) HRC-WHU Dataset: The high-resolution cloud cover validation dataset consists of 150 high-resolution images collected from Google Earth (Google Inc.) with the resolution ranging from 0.5 to 15 m. It covers different global regions, including barren, vegetation, urban, snow/ice, and water. These images include RGB channels and are labeled with clouds and clear sky. In our experiments, we divide them into the training and test sets in the ratio of 3:1 to verify the effectiveness of our method on high-resolution satellite images.

5) GF-1 WFV Cloud and Cloud Shadow Cover Validation Data: The GF-1 dataset includes 108 Level-2A scenes from the WFV imaging system of the GaoFen-1 satellite and covers different global land-cover types with various cloud conditions. These images have a spatial resolution of 16 m and are comprised of four spectral bands, including visible and NIR spectral regions. The reference masks of these images label cloud, cloud shadow, and non-cloud regions. To evaluate the effectiveness of our method for cloud shadow detection, we divide them into 42, 38, and 28 scenes for training, validation, and testing, respectively.

B. Implementations

1) Network Structure: The full convolution-based encoder–decoder model is utilized to implement end-to-end cloud detection, as shown in Fig. 3. The network with the output stride of eight fuses multiscale features in the concatenated manner by skipping connection. The decoder realizes layer-by-layer upsampling of features with the bilinear interpolation and outputs pixel-level prediction results. The PPM [39] is used to make the teacher network achieve better performance. The difference between the teacher and student networks lies in the choice of backbone network. The teacher network uses dilated ResNet50 as the backbone network. Meanwhile, we validate our distillation method with different lightweight network structures as student networks, namely, ResNet18 [48] with the half number of channels (denoted ResNet18-h) and MobilenetV2 [16]. Halving the number of channels in ResNet18 reduces the model parameters by a factor of 3.3 to only 3.29M. MobilenetV2 designs the inverted residual block to get better performance with the reduced computation and has only 2.31M parameters in our task.

2) Training Setting: The teacher network and MobilenetV2 both use the network pre-trained on ImageNet [49], and ResNet18-h is trained from scratch. When the network is fed with multiple bands, we load the pre-trained weights of the

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TABLE II
ABLATION EXPERIMENTAL RESULTS OF THE PROPOSED DISTILLATION METHOD ON THE LANDSAT 8 DATASET

Method	IoU(%)			mIoU(%)	OA(%)	F1(%)	UA(%)	PA(%)	Params(M)	FLOPs(G)
	thick_c	thin_c	non_c							
T:ResNet50	85.62	73.44	92.47	83.84	92.51	91.01	91.30	90.73	53.20	232.79
S1:ResNet18-h	76.97	63.03	89.70	76.57	88.82	86.29	86.19	86.41	3.29	14.64
S1+PAD	79.10	66.63	91.16	78.97	90.16	87.89	87.82	88.01	3.29	14.64
S1+CCD	80.21	67.21	90.85	79.42	90.30	88.20	88.05	88.44	3.29	14.64
S1+PAD+CCD	81.94	69.04	91.50	80.83	91.04	89.11	89.26	88.96	3.29	14.64
S2:MobileNetV2	80.84	67.24	90.98	79.69	90.45	88.36	88.39	88.35	2.31	4.63
S2+PAD	82.81	69.99	91.58	81.46	91.35	89.52	89.49	89.57	2.31	4.63
S2+CCD	84.09	71.60	92.06	82.58	91.92	90.22	90.27	90.20	2.31	4.63
S2+PAD+CCD	85.20	72.55	92.20	83.32	92.26	90.68	90.75	90.62	2.31	4.63

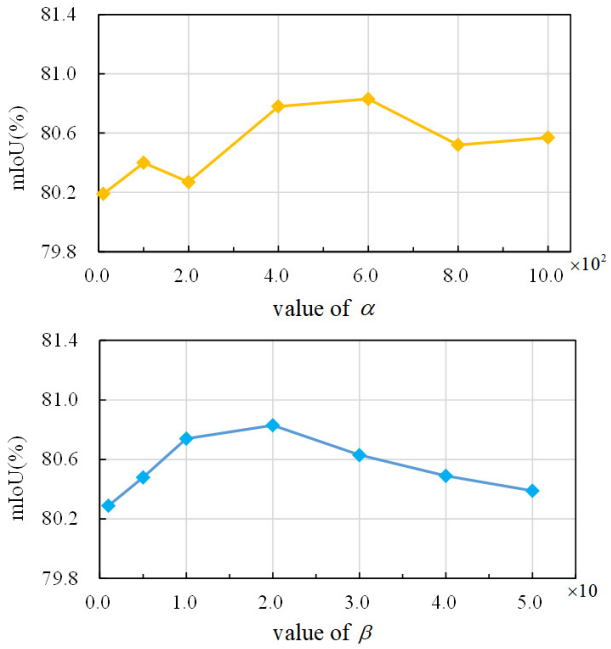

Fig. 8. Hyperparameter sensitivity study of α and β with ResNet18-h on the Landsat 8 dataset.

RGB data and initialize the weights of the other bands with the normal distribution. Following the previous works [25], [38], the loss weights α and β are set to 600 and 20 to maintain a balance with the cross-entropy loss. The stochastic gradient descent (SGD) as the optimizer is used for training the network with a momentum of 0.9. Our networks are trained for 300 epochs with a batch of eight images. The initial learning rate is set to 0.002, and the "poly" learning rate update strategy [50] is used with a power of 0.9 and a weight decay of 0.0001. Some general data augmentations are implemented, including random rotation, random horizontal, and random vertical flipping. Our experiments are performed with the PyTorch 1.3 framework on an RTX-2080Ti GPU.

3) Evaluation Metrics: We quantitatively evaluate cloud detection results using common accuracy assessment metrics, including mean pixel Intersection-over-Union (mIoU), overall pixel accuracy (OA), $F1$ score, recall or user's accuracy (UA), and precision or producer's accuracy (PA), respectively. They are calculated in detail as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{k=1}^{C}\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{15}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{16}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{17}$$

$$\text{presion} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{18}$$

$$F_1 = \left(1+\beta^2\right)\cdot\frac{\text{presion}\cdot\text{recall}}{\beta^2\cdot\text{presion}+\text{recall}},\quad \beta = 1 \tag{19}$$

in which TP, TN, FP, and FN are the number of true positives, true negatives, false positives, and false negatives, respectively.

To compare the efficiency of the different methods, we evaluate the number of parameters of the model (denoted as Params) and the number of FLOPs for the model to compute a sliced image with 512 × 512 pixels. In addition, the inference time (denoted as Infer time) of each model is also evaluated at an image with original resolution and a sliced image with 512 × 512 pixels.

C. Ablation Experiments

We first validate the upgrading of the baseline by the different distillation components from the quantitative and qualitative aspects. Afterward, we analyze the effectiveness of the two distillation components individually.

1) Quantitative Results: To verify the effectiveness of our feature-level (CCD) and pixel-level (PAD) distillation methods, respectively, we sequentially add distillation components for experiments. We report the results of the distillation module on two student networks in Table II. Under the knowledge transfer of the CCD and PAD, the mIoU metrics of ResNet18-h and MobileNetV2 are increased by 4.26% and 3.63%, respectively. In particular, the improvement for thin and thick clouds in detection accuracy is more obvious. For MobileNetV2, thick and thin clouds are promoted by 4.36% and 5.31% mIoU, respectively. The gap for thin clouds between MobileNetV2 and the teacher is reduced from 6.20% to 0.89%. This confirms the effect of our distillation

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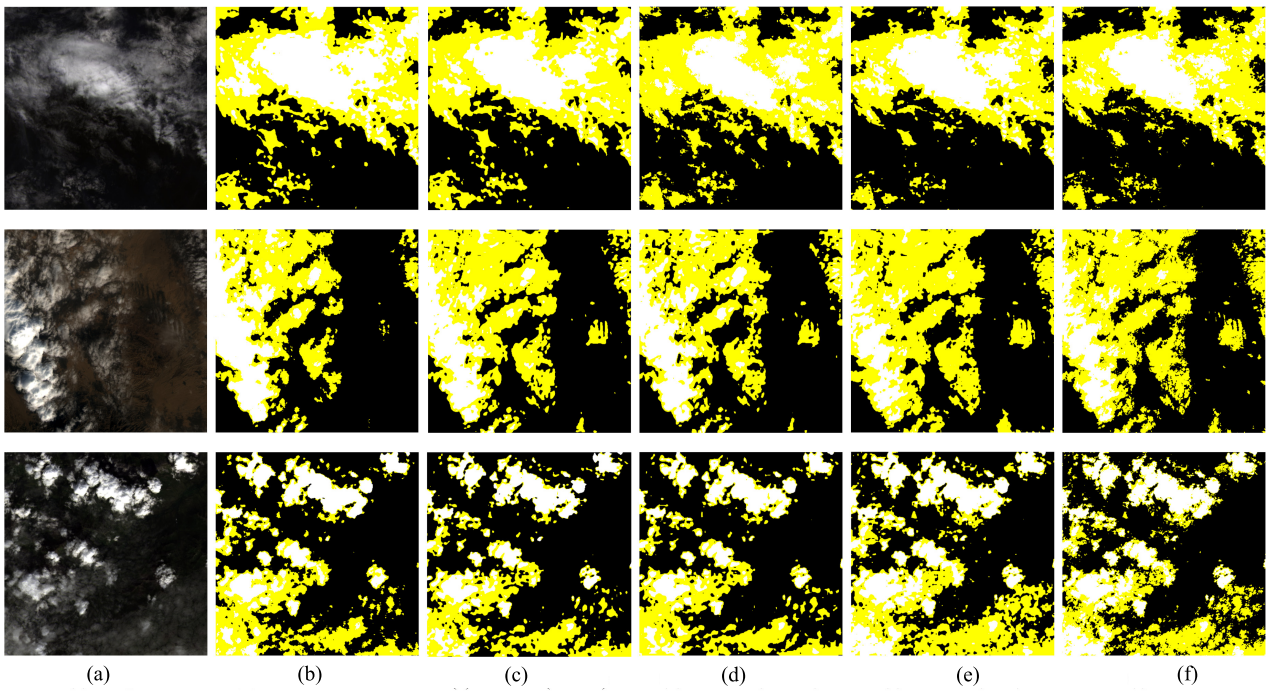

Fig. 9. Predicted results of the different distillation components on the Landsat 8 dataset. (a) Image. (b) Results of ResNet18-h. (c) Results of ResNet18-h with PAD. (d) Results of ResNet18-h with CCD. (e) Results of ResNet18-h adding the PAD and CCD. (f) Label. The white, yellow, and black colors in the masks denote thick clouds, thin clouds, and clear sky, respectively.

method for hard categories. Also, the combined application of CCD and PAD brings the maximum benefit to the network. In particular, feature-level knowledge has more powerful outcomes than pixel level, reflecting the importance of feature representation in KD for segmentation networks. Notably, MobileNetV2 achieves detection accuracy comparable to the teacher network with a huge reduction in the model parameters and FLOPs. Furthermore, we analyze the sensitivity of the loss weights hyperparameters α and β to the model through several experiments. Fig. 8 shows the robustness of each hyperparameter.

2) Qualitative Results: The visualization results of cloud detection on sliced images by adding distillation components are shown in Fig. 9. In Fig. 9(c), the PAD module distills the pixel-level predicted output from the teacher, allowing the student to capture inconspicuous thin cloud pixels. Fig. 9(d) demonstrates that the CCD enhances the performance of thin cloud detection against complex backgrounds by distilling the feature category correlation. The student network is able to capture smaller thin cloud fragments more accurately and has a more detailed prediction of thin cloud boundaries. This validates that the CCD is effective in helping the student network capture the common characteristics of thin clouds and detecting thin clouds that are not easily visible in the background. As shown in Fig. 9(e), the predictions of the student network are effectively improved with our distillation framework, especially for inconspicuous thin clouds and irregular cloud boundaries compared with the result before distillation.

3) Analysis of Feature CCD: The proposed CCD optimizes the feature representation of student networks by distilling the feature CRC. From Table II, the CCD for feature-level distillation brings 2.85% and 2.89% mIoU to ResNet18-h and MobileNetV2, respectively. Especially for thick and thin clouds, in the example of ResNet18-h, the CCD improves by 3.24% and 4.18% mIoU, respectively.

TABLE III
EFFECTIVENESS OF NORMALIZING THE FEATURE SIMILARITY COMPARED WITH DIRECT DISTILLATION ON THE LANDSAT 8 DATASET

Method	CosSim	CRC	mIoU(%)	OA(%)	F1(%)
S1:ResNet18-h	✓		78.50	89.85	87.58
		✓	79.42	90.30	88.20
S2:MobileNetV2	✓		82.25	91.76	90.02
		✓	82.58	91.92	90.22

In our CCD, normalizing the feature similarity is necessary to eliminate the gap with different feature depths. Previous feature distillations directly match the similarity of pairs of pixels and usually use $L2$ loss. The results in Table III report the effectiveness of our CRC map. CosSim denotes the distillation of the feature category cosine similarity using $L2$ loss. Our CCD calculates the normalized CRC distilled by KL divergence and yields higher gains for the knowledge transfer.

To prove the improvement of CCD on the feature map, we visualize the feature CRC map for thin clouds of different networks in Fig. 10. The feature CRC is constructed on the similarity mapping between pixels and different class regions. It is well known that high intraclass compactness and interclass dispersion of the feature representation can lead to better segmentation results. Compared with the teacher network in

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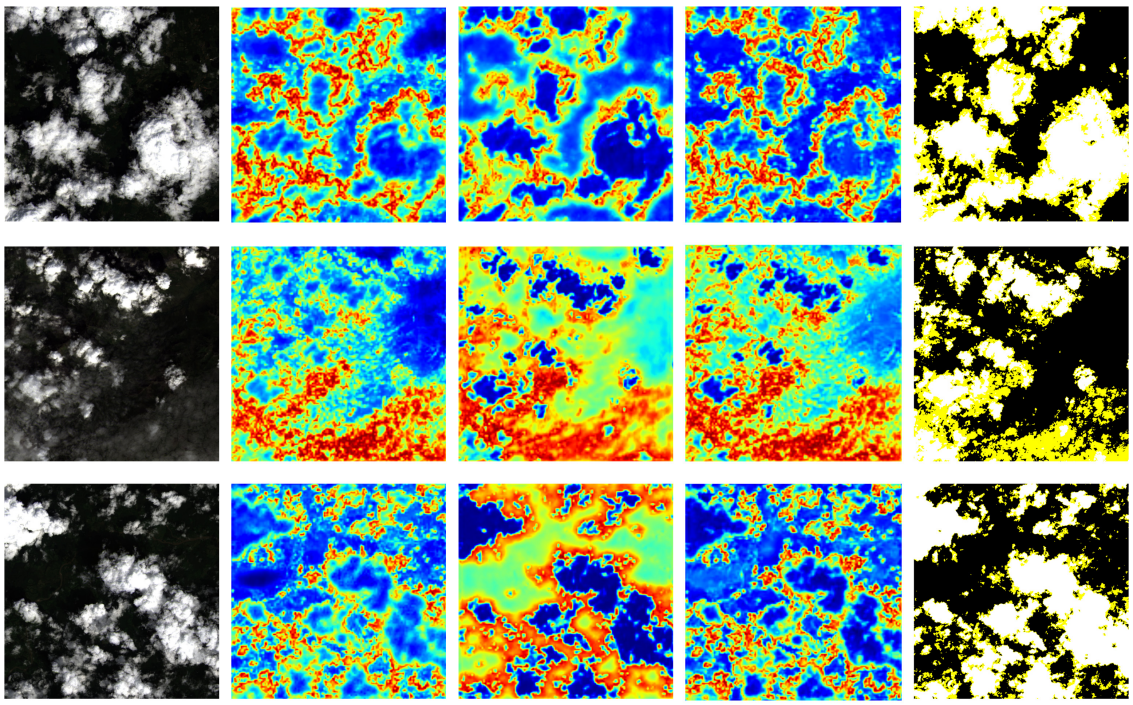

Fig. 10. Visualization of thin cloud CRC map for different networks on the Landsat 8 dataset. (a) Image. (b) Thin cloud CRC map of the teacher network. (c) Thin cloud CRC map of ResNet18-h as the student network. (d) Thin cloud CRC map of ResNet18-h with CCD. (e) Label. The white, yellow, and black colors in the masks denote thick clouds, thin clouds, and clear sky, respectively.

TABLE IV
EXPERIMENTAL RESULTS USING PAD AND KD ON THE LANDSAT 8 DATASET

Method	KD	PAD	mIoU(%)	OA(%)	F1(%)
S1:ResNet18-h			76.57	88.82	86.29
	✓		78.05	89.70	87.28
		✓	78.97	90.16	87.89
S2:MobileNetV2			79.69	90.45	88.36
	✓		80.45	90.86	88.86
		✓	81.46	91.35	89.52

TABLE V
EXPERIMENTAL RESULTS COMPARED WITH OTHER ADVANCED DISTILLATION METHODS ON THE LANDSAT 8 DATASET

Method	mIoU(%)	OA(%)	F1(%)
T:ResNet50	83.84	92.51	91.01
S:ResNet18-h	76.57	88.82	86.29
S + KD [18]	78.05	89.70	87.28
S + FitNet [20]	78.49	89.98	87.57
S + Pair-wise [24]	79.26	90.28	88.08
S + ours	80.83	91.04	89.11

Fig. 10(b), the CRC map of ResNet18-h as the student network in Fig. 10(c) has relatively high confidence values with the background region features. As expected, the inconspicuous target characteristic of thin clouds can result in thin cloud pixels being easily confused with the background, making thin clouds less detectable. In contrast, in the case with CCD in Fig. 10(d), the feature intraclass consistency description of thin clouds is significantly improved. The feature similarity of the thin clouds and the background is clearly decreased on most background pixels. The student network's feature descriptions of thin clouds are refined by learning more intensive feature category correlation from the teacher. In addition, we notice that the original student network embodies a high category feature difference between thin and thick clouds, while the feature difference is weakened after CCD. We consider the local relevance change acceptable, because the category feature relationships of the student network have been improved over the whole image, which is more important for prediction.

4) Analysis of PAD: The PAD module fixes the incorrect pixel knowledge of the teacher network and avoids misdirection on pixels that are hard to predict. We compare the direct pixel-level imitation [18] (denoted as KD) method to verify the effect of the PAD module. Table IV shows that the PAD module enables pixel-level distillation to improve by 0.92% and 1.01% mIoU on two student networks, respectively, compared with KD. Thus, it is favorable to reweight the pixel-level distillation loss with the predicted confidence of the teacher.

D. Comparison With State-of-the-Art Methods

We compare our distillation method with other distillation methods on the Landsat 8 dataset. We also exhibit

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TABLE VI
COMPARISON EXPERIMENTAL RESULTS WITH OTHER CLOUD DETECTION METHODS ON THE LANDSAT 8 DATASET

Method	IoU(%)			mIoU(%)	OA(%)	F1(%)	UA(%)	PA(%)	Params(M)	FLOPs(G)
	thick_c	thin_c	non_c							
L-Unet [35]	68.67	53.20	87.64	69.84	85.26	81.43	81.04	81.86	0.12	6.79
ENet [51]	74.36	58.26	88.83	73.82	87.37	84.34	84.62	84.12	0.36	2.29
Deeplabv3 [52]	76.24	62.48	88.39	75.70	88.22	85.75	85.79	85.72	39.05	47.93
CDNet [5]	80.17	65.05	89.46	78.23	89.44	87.42	87.87	87.02	77.40	387.43
PSPNet [39]	82.11	68.84	90.84	80.60	90.83	88.98	89.04	88.93	70.42	262.23
MFCNN [4]	83.17	70.63	91.81	81.87	91.59	89.78	89.50	90.09	14.78	126.15
ResNet18-h (ours)	81.69	69.04	91.50	80.83	91.04	89.11	89.26	88.96	3.29	14.64
MobileNetV2 (ours)	85.20	72.55	92.20	83.32	92.26	90.68	90.75	90.62	2.31	4.63

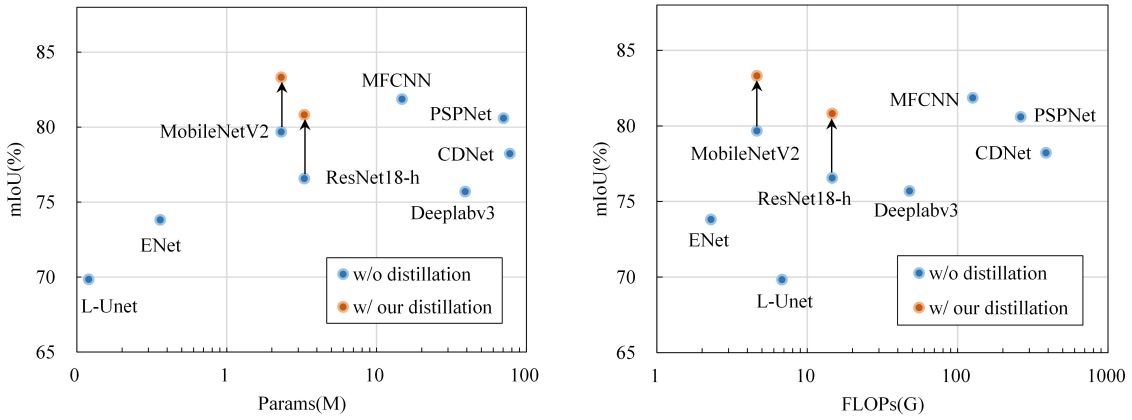

Fig. 11. Comparisons in terms of parameters, FLOPs, and mIoU for different models on the Landsat 8 dataset.

the comparison between lightweight student networks with our distillation method and other advanced cloud detection methods.

1) Comparison With Other Distillation Methods: We compare the proposed distillation method with distillation methods: KD, FitNet, and pairwise distillation. KD proposed by Hinton is used to directly distill the pixel-level score map of networks. FitNet is proposed in the classification task to distill the intermediate layer representation of the teacher as a hint. Also, additional parameters are introduced to align the spatial scale and the depth of the middle layer between networks. Following [20], we use 1×1 convolution to align feature channels of the student with the teacher in the FitNet experiment. Pairwise distillation is proposed in [24] for the image segmentation of natural scenes, which calculates the pairwise similarity of pixels to build an affinity graph. In particular, we calculate the global pixel affinity of the whole image, and the pixelwise distillation is also be applied.

Table V demonstrates that our method approach is more effective than other distillation strategies. KD effectively improves the performance but with limited gain. Since FitNet directly matches the middle feature of networks, it ignores the feature representation capability with different depths, resulting in a smaller accuracy gain. The pixel pairwise similarity relationship of the pairwise distillation is dense, while our approach unites the category label of pixels to build a kind of compact pixel-category relationship, which is more suitable for the cloud detection task.

TABLE VII
EXPERIMENTAL RESULTS OF OUR METHOD WITH DIFFERENT BANDS

Method	Bands	mIoU(%)	OA(%)	F1(%)
ResNet18-h (ours)	2-4	80.83	91.04	89.11
MobileNetV2 (ours)		83.32	92.26	90.68
ResNet18-h (ours)	2-6,10	82.25	91.76	90.01
MobileNetV2 (ours)		84.68	92.91	91.52
ResNet18-h (ours)	All	83.70	92.39	90.91
MobileNetV2 (ours)		85.14	93.17	91.80

2) Cloud Detection Results of Landsat 8 Dataset: We also make a comparison with other advanced cloud detection networks on Landsat 8 satellite images. Our comparison methods include classical semantic segmentation networks, such as ENet, DeeplabV3, and PSPNet, where ENet is a lightweight real-time semantic segmentation network. Other cloud detection networks, such as L-Unet, CDnet, and multiscale features-convolutional neural network (MFCNN), are also used for comparison, where L-unet is a lightweight cloud detection network. Similar to our work, MFCNN is also proposed to detect thin and thick clouds.

We consider the detection accuracy, the number of parameters, and the amount of computations (i.e., FLOPs) of the network model to comprehensively compare different cloud

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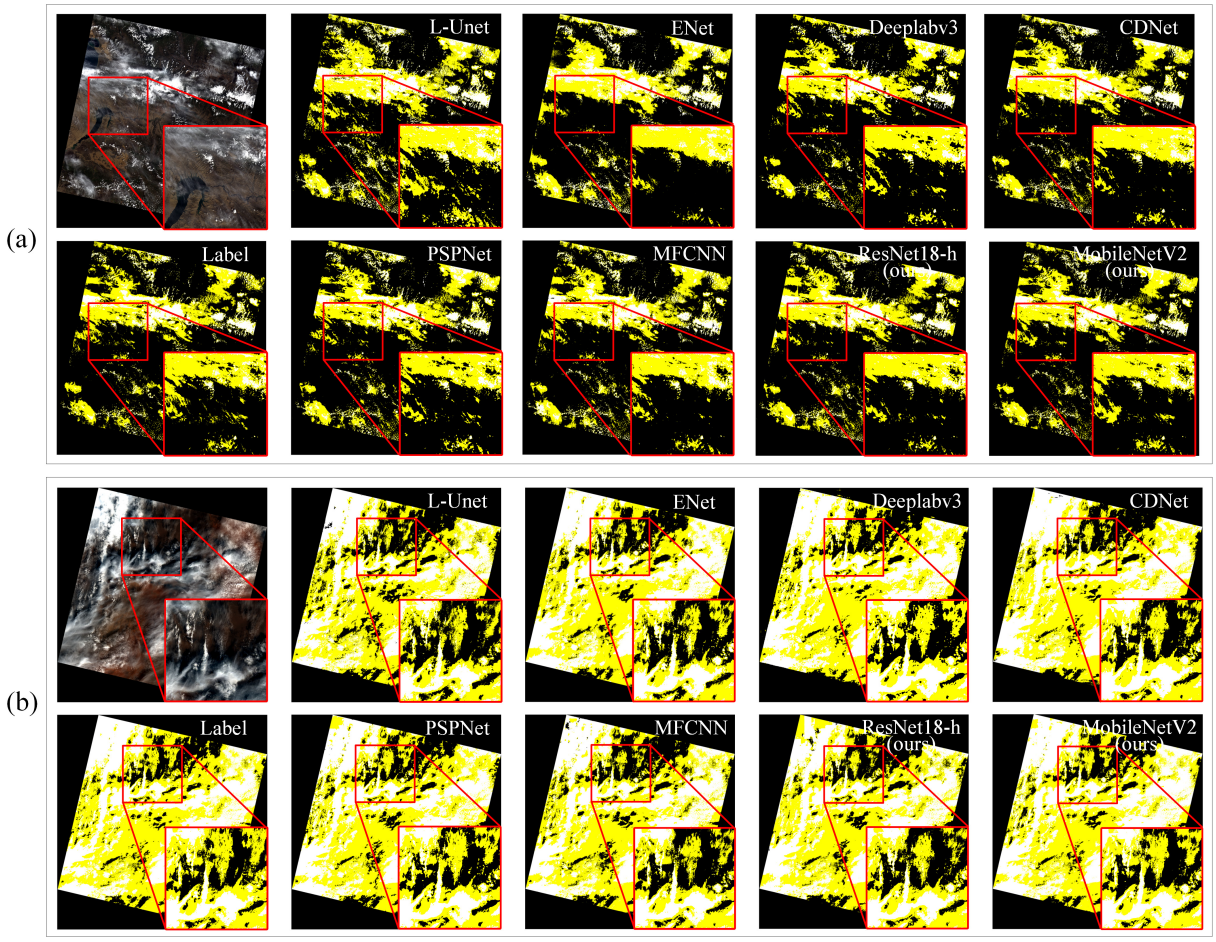

Fig. 12. Visualization of prediction results for Landsat 8 images. (a) Grass/crops area (scene ID: LC81320352013243LGN00). (b) Barren area (scene ID: LC81990402014267LGN00). The white, yellow, and black colors in the masks denote thick clouds, thin clouds, and clear sky, respectively.

detection methods. In Table VI, MobileNetV2 with our distillation achieves the highest accuracy with 83.32% mIoU among all networks. Also, the performance of student networks is further improved by inputting multiband images. For the efficiency of networks, both L-Unet and ENet use lightweight networks with fewer parameters and FLOPs, but the detection accuracy achieved is unsatisfactory. Compared with cumbersome models Deeplabv3, PSPNet, CDNet, and MFCNN, our distillation method allows the lightweight MobileNetV2 to outperform these models with only 16% model parameters and 4% FLOPs of MFCNN that performs best in cumbersome models. Fig. 11 shows the accuracy and efficiency of different cloud detection methods. Our method effectively improves the accuracy of the lightweight network, and the red dots indicate lightweight models with our approach. Also, MobileNetV2 achieves the highest accuracy with lesser parameters and complexity.

The accuracy of cloud detection is also influenced by the input spectral bands. To compare the performance of our model under different band conditions, the inputs of RGB, NIR, SWIR, and TIR bands (2–6 and 10) and all available bands for our model are evaluated. The results are given in Table VII, and the accuracy evaluation results for models indicate that the input of all available bands achieves the highest mIoU in the Landsat 8 test set. The addition of infrared bands (5, 6, and 10) further improves the accuracy of models compared with using only RGB bands (2–4). Therefore, our method is effective at different band settings, and the more the spectral bands, the greater the accuracy gain.

The visualization results for Landsat 8 images are presented in Fig. 12. In different scenarios, compact networks (ResNet18-h and MobileNetV2) using our distillation method show accurate detection at thin clouds, broken clouds, and cloud boundaries. Compared with lightweight networks, our MobileNetV2 model captures thin clouds that are not easily noticeable. Our model achieves comparable and competitive detection results compared with the DCNN-based approaches.

3) Cloud Detection Results of Landsat 7 Dataset: We also validate the validity of our distillation method on Landsat 7 satellite images. We compare the same cloud detection method as the Landsat 8 dataset. We train and test different networks using bands 1–4 images (RGB and NIR). In addition, we compare with the direct pixel-level imitation [18] (denoted as KD).

Table VIII reports the predicted results of the different methods. We use our distillation method for the ResNet18-h and MobileNetV2 student networks on the Landsat 7 dataset to bring 4.07% and 2.91% improvement in mIoU metrics that is superior to KD. The accuracy of the thin clouds has

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TABLE VIII
COMPARISON EXPERIMENTAL RESULTS WITH OTHER CLOUD DETECTION METHODS ON THE LANDSAT 7 DATASET

Method	**IoU(%)**			**mIoU(%)**	**OA(%)**	**F1(%)**	**UA(%)**	**PA(%)**	**Params(M)**	**FLOPs(G)**
	thick_c	**thin_c**	**non_c**							
L-Unet [35]	81.23	39.51	92.18	70.98	91.21	80.74	79.43	83.06	0.12	6.83
Deeplabv3 [52]	77.66	48.24	91.13	72.34	90.90	82.62	81.52	83.89	39.05	48.14
ENet [51]	82.35	45.62	92.47	73.48	91.73	83.02	82.44	83.76	0.36	2.29
CDNet [5]	83.95	49.46	93.06	75.49	92.49	84.62	83.68	85.71	77.40	387.47
PSPNet [39]	83.56	52.70	93.07	76.44	92.66	85.49	84.93	86.11	70.42	262.23
MFCNN [4]	85.24	50.72	93.95	76.64	93.19	85.41	84.21	87.18	14.78	126.18
ResNet18-h [48]	82.29	48.12	92.53	74.31	92.09	83.79	82.16	85.82	3.29	14.75
ResNet18-h (KD)	84.71	51.61	92.99	76.43	92.57	85.39	85.44	85.34	3.29	14.75
ResNet18-h (ours)	84.34	55.42	93.77	78.38	93.40	86.85	86.40	87.32	3.29	14.75
MobileNetV2 [16]	84.34	50.34	93.41	76.03	92.76	85.02	84.10	86.13	2.31	4.66
MobileNetV2 (KD)	85.44	53.19	93.71	77.45	93.22	86.11	85.23	87.18	2.31	4.66
MobileNetV2 (ours)	86.58	56.52	94.06	79.06	93.68	87.32	86.74	87.94	2.31	4.66

TABLE IX
COMPARISON EXPERIMENTAL RESULTS WITH OTHER CLOUD DETECTION METHODS ON THE AIR-CD DATASET

Method	mIoU(%)	OA(%)	F1(%)	UA(%)	PA(%)
Deeplabv3 [52]	88.04	95.68	93.51	93.45	93.58
MFCNN [4]	89.50	96.21	94.36	94.31	94.41
PSPNet [39]	90.04	96.43	94.67	94.64	94.70
CDNet [5]	90.86	96.74	95.13	95.07	95.19
DABNet [36]	91.40	96.92	95.44	95.38	95.50
ResNet18-h (ours)	90.46	96.56	94.91	94.96	94.86
MobileNetV2 (ours)	91.42	96.95	95.45	95.39	95.51

improved significantly with 7.3% and 5.9% mIoU. Compared with other cloud detection methods, our approach also allows both student network to achieve optimal performance with lesser parameters and FLOPs.

4) Cloud Detection Results of High-Resolution RSIs: To verify the generalization performance of our method, we conduct experiments on the high-resolution satellite dataset AIR-CD and HRC-WHU. Table IX shows that MobileNetV2 with our CAKD framework achieves 91.42% in mIoU and 95.45% in $F1$ score on the AIR-CD dataset, which outperforms other cloud detection methods.

The accuracy comparison results of different methods on HRC-WHU dataset are reported in Table X. Our method results in 2.26% and 1.74% improvement in mIoU for ResNet18-h and MobileNetV2, respectively. The MobileNetV2 with our CAKD achieves the mIoU accuracy of 89.31%, which is higher than other cloud detection networks. Cloud detection examples on high-resolution satellite images of HRC-WHU dataset are shown in Fig. 13, with scenes from snow/ice, urban, and water areas, respectively. It is observed that the MobileNetV2 baseline method tends to incorrectly detect ground objects. Also, the MobileNetV2 with our CAKD yields relatively more accurate cloud detection results and reduces noticeable mistakes in local regions.

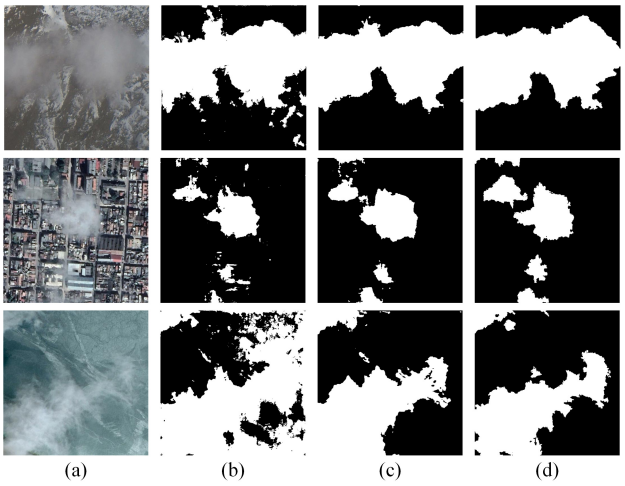

Fig. 13. Visualization of cloud detection results for the HRC-WHU dataset. (a) Images. (b) Results of MobileNetV2. (c) Results of MobileNetV2 with our CAKD. (d) Label.

TABLE X
COMPARISON EXPERIMENTAL RESULTS WITH OTHER CLOUD DETECTION METHODS ON THE HRC-WHU DATASET

Method	mIoU(%)	OA(%)	F1(%)	UA(%)	PA(%)
Deeplabv3 [52]	85.90	92.62	92.40	92.32	92.49
PSPNet [39]	87.46	93.45	93.30	93.54	93.30
MFCNN [4]	88.14	93.84	93.69	93.82	93.57
CDNet [5]	88.52	94.06	93.90	93.93	93.88
ResNet18-h [48]	84.71	91.95	91.70	91.60	91.82
ResNet18-h (ours)	86.97	93.23	93.02	92.87	93.18
MobileNetV2 [16]	87.57	93.51	93.36	93.56	93.20
MobileNetV2 (ours)	89.31	94.50	94.34	94.30	94.39

5) Cloud and Cloud Shadow Detection Results of GF-1 Dataset: To verify the applicability of our method for cloud and cloud shadow detection, we apply our distillation method

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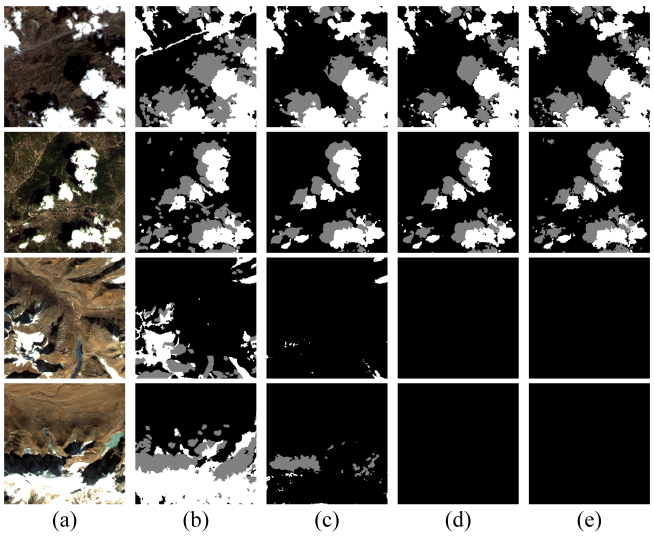

Fig. 14. Visualization of cloud detection results for GF-1 dataset. (a) Images. (b) Results of MFC method. (c) Results of MobileNetV2. (d) Results of MobileNetV2 with our CAKD. (e) Label. The white, gray, and black colors in the masks denote cloud, cloud shadow, and clear sky, respectively.

TABLE XI
COMPARISON EXPERIMENTAL RESULTS WITH OTHER CLOUD DETECTION METHODS ON THE GF-1 DATASET

Method	mIoU(%)	OA(%)	F1(%)	UA(%)	PA(%)
MFC [30]	80.64	95.97	88.22	86.75	89.90
Deeplabv3 [52]	82.46	96.49	89.47	87.33	92.13
PSPNet [39]	83.07	96.51	89.95	88.48	91.65
MFCNN [4]	83.68	96.63	90.39	89.45	91.43
CDNet [5]	84.09	96.93	90.58	88.71	92.79
ResNet18-h [48]	80.88	96.12	88.28	87.03	89.71
ResNet18-h (ours)	83.28	96.67	90.04	88.51	91.86
MobileNetV2 [16]	83.04	96.66	89.85	88.36	91.67
MobileNetV2 (ours)	85.17	97.15	91.33	89.84	93.03

on the GF-1 dataset. Table XI reports the accuracy evaluation results of the different methods, in which the multifeature combined (MFC) method is the baseline method for the GF-1 WFV dataset. On the one hand, our distillation method is effective when applied on ResNet18-h and MobileNetV2, and their mIoU is improved by 2.4% and 2.13%, respectively. On the other hand, our MobileNetV2 outperforms other cloud detection methods, and the accuracy is significantly improved compared with the traditional MFC method.

Fig. 14 shows some examples of cloud detection results on the GF-1 dataset. The first two rows show that on cloud shadow detection, the MobileNetV2 with our CAKD in Fig. 14(d) improves the inaccurate boundary results of the baseline method in Fig. 14(c). Moreover, our results have fewer false detections of cloud shadows compared with the MFC method in Fig. 14(b). The last two rows of Fig. 14 show the detection results in the snow and ice scenario. We can see that the MFC method easily mistakes the snow/ice and mountain shadows as clouds and cloud shadows, and the MobileNetV2 after applying our CAKD can better distinguish snow/ice, clouds, and cloud shadows. We analyze the effectiveness of our approach in that through our CAKD, the student network enhances the category-consistent representation of cloud and cloud-shadow regions, which indicates that our CAKD is also applicable to cloud shadow detection.

TABLE XII
COMPARISON OF THE INFER TIME OF DIFFERENT METHODS. *THE SUBIMAGES USED FOR THE EXPERIMENTS ARE FROM THE LANDSAT 8 DATASET

Method	Params(M)	FLOPs(G)	Infer time(s)		
			Landsat-8	GF-1	Sub-image*
MFC [30]	N/A	N/A	/	278.6	/
Fmask [29]	N/A	N/A	109.3	/	/
L-Unet [35]	0.12	6.79	19.2	83.3	0.077
ENet [51]	0.36	2.29	21.5	89.7	0.082
Deeplabv3 [52]	39.05	47.93	22.8	105.3	0.105
PSPNet [39]	70.42	262.23	40.5	174.7	0.168
MFCNN [4]	14.78	126.15	22.8	101.8	0.097
CDNet [5]	77.40	387.43	53.1	235.1	0.224
ResNet18-h (ours)	3.29	14.64	**17.4**	**82.5**	**0.075**
MobileNetV2 (ours)	2.31	4.63	18.7	88.4	0.081

E. Time Efficiency

In this section, we evaluate the time efficiency of different cloud detection methods. In particular, we test the Infer time for images of varying sizes. In our implementation, we randomly select two RSIs from Landsat 8 and Gaofen-1 satellites with the sizes of 7581 × 7751 pixels and 17116 × 15309 pixels, respectively. The baseline method Fmask for Landsat 8 and the MFC for GF-1 are performed in CPU mode with an Intel Core i7-7700, while the DCNN-based method is tested with an NVIDIA Tesla-P100 GPU. The DCNN-based method performs inference based on the subimages of 512 × 512 pixels and then stitches the results of subimages to get the final result of the whole image. Thus, we also report the time to predict a subimage for the DCNN-based method.

Table XII lists the parameters, FLOPs, and Infer time of different methods. The metrics of the DCNN-based methods are tested with three bands as input. We test 100 sliced subimages from the Landsat 8 dataset and report the average Infer time. It is evident that the DCNN-based methods cost less time than the traditional Fmask and MFC method due to the efficient computation of GPU. The Infer time of our ResNet18-h model is 17.4, 82.5, and 0.075 s for Landsat 8, Gaofen-1 images, and a subimage, which is shorter than other methods. L-Unet and ENet have comparable Infer time to ResNet18-h, but their cloud detection accuracy is inferior (as shown in Section IV-D). We also observe that although MobileNetV2 has fewer parameters and FLOPs, it is slightly slower than ResNet18-h due to its special depthwise separable convolution. However, we consider it acceptable that the detection accuracy of MobileNetV2 is better than other models. Comprehensively, compared with other cloud detection networks, our ResNet18-h and MobileNetV2 models achieve satisfactory accuracy with relatively short Infer time.

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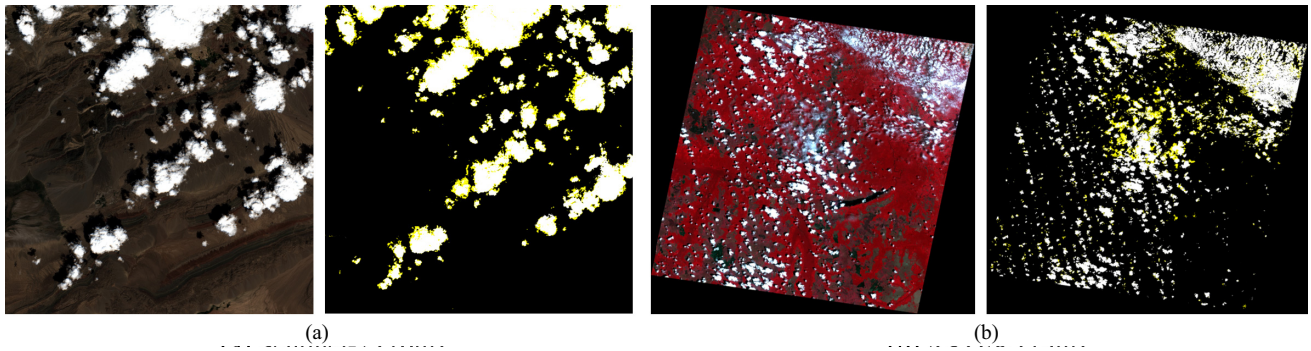

Fig. 15. Cloud detection examples of our MobileNetV2 on other satellite images. The white, yellow, and black colors in the masks denote thick clouds, thin clouds, and clear sky, respectively. (a) Sentinel-2 (10 m). (b) ASTER (15 m).

TABLE XIII
DISTILL DIFFERENT MODELS WITH OUR METHOD ON THE LANDSAT 8 DATASET

Model	method	mIoU(%)	OA(%)	F1(%)	Params(M)	FLOPs(G)
PSPNet [39]	tea	80.60	90.83	88.98	70.42	262.23
	stu	74.07	87.38	84.59	3.16	12.65
	stu (ours)	77.40	89.21	86.86	3.16	12.65
MFCNN [4]	tea	81.87	91.59	89.78	14.78	126.15
	stu	76.20	88.72	86.02	3.55	31.62
	stu (ours)	79.55	90.49	88.26	3.55	31.62

F. Discussion

1) Portability: In Section IV-D, we trained and tested our cloud detection models on various datasets to demonstrate the effectiveness of our approach, illustrating the scalability of our CAKD for cloud detection under different satellites, different resolutions, and different spectral settings. In addition, we test our model on other satellites without training on them to verify the portability of our model. Our MobileNetV2 model trained on the Landsat 8 satellite samples labeled with thin clouds and thick clouds is applied to cloud detection on other satellite images. The test images are the visible imagery of the Sentinel-2 satellite and the visible-to-near infared (VNIR) imagery of the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) satellite sensor, respectively. As shown in Fig. 15, our model generates satisfactory cloud mask results on the Sentinel-2 images with a resolution of 10 m and the ASTER satellite images with a resolution of 15 m. We attribute the effectiveness to the fact that clouds have similar visual characteristics on different training images (Landsat 8) and test images (Sentinel-2 or ASTER). Moreover, our model is trained on globally different land-cover types, and the training data are diverse.

2) Versatility: To prove the versatility of our CAKD, we apply our method on different DCNN-based cloud detection models. The distillation results on Landsat8 dataset are shown in Table XIII. We conduct experiments with PSPNet and MFCNN model as the teacher model. For PSPNet, we replace the backbone of the teacher network (ResNet101) with ResNet18-h as the lightweight student network. For the MFCNN, we take the MFCNN with the half number of channels as the student network. It can be seen that our method achieved excellent performance for different types of cloud detection models.

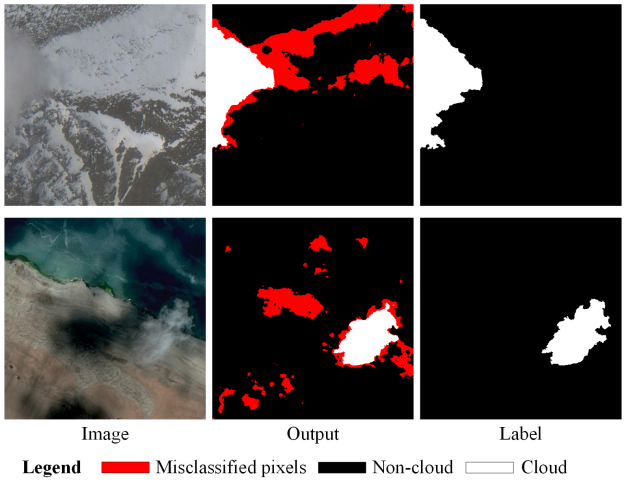

Fig. 16. Cloud detection errors of the teacher model on high-resolution RSIs of the HRC-WHU dataset.

3) Limitations: Although the proposed CAKD framework can improve the lightweight model's accuracy in complex scenes, there are still a few situations that are difficult to cope with for cloud detection of high-resolution satellite images. The results of the sample on the HRC-WHU dataset in Fig. 13 show that our approach can enhance the detection results in some complex scenes of the lightweight model. However, since the cloud detection task for high-resolution RSIs is susceptible to ground objects and illumination, our teacher model may produce prediction errors in complex scenarios, as shown in Fig. 16. The obvious mistakes of the teacher model can affect the outcome of the KD. In the future, we will optimize the performance of complicated teacher model to improve the robustness of lightweight model on high-resolution RSIs.

V. CONCLUSION

In this article, we present an effective CAKD framework tailored for the compact cloud detection model. The proposed CRC module measures the pixel-category correlation of the

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feature map of networks. The CCD enables the compact network to learn the coherent common features of clouds and reinforce feature differences between clouds and cloud-like objects through distilling the CRC knowledge. The PAD module extracts the teacher's pixel prediction value to reweight the distillation loss for adaptive distillation, which mitigates the negative guidance from the teacher network. We validate the effectiveness and universality of our distillation method on the Landsat 8, Landsat 7, Gaofen-2, Gaofen-1, and Google Earth datasets. Our approach is better suited to the cloud detection task than other distillation methods and allows lightweight MobileNetV2 to outperform prevalent cloud detection methods with better efficiency. Concerning future work, we intend to optimize the performance of complicated teacher model to further improve the accuracy of the lightweight network in complex scenes of high-resolution RSIs. In addition, we expect to leverage rich spectral bands captured by multispectral or hyperspectral satellites to boost cloud detection results in complicated scenarios.

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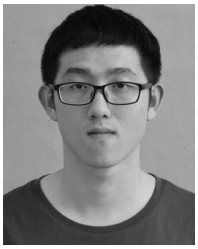

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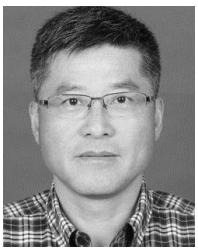

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